



Chapter4

Discrete-time Systems





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Classification of Discrete Systems

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LTI Discrete-time Systems

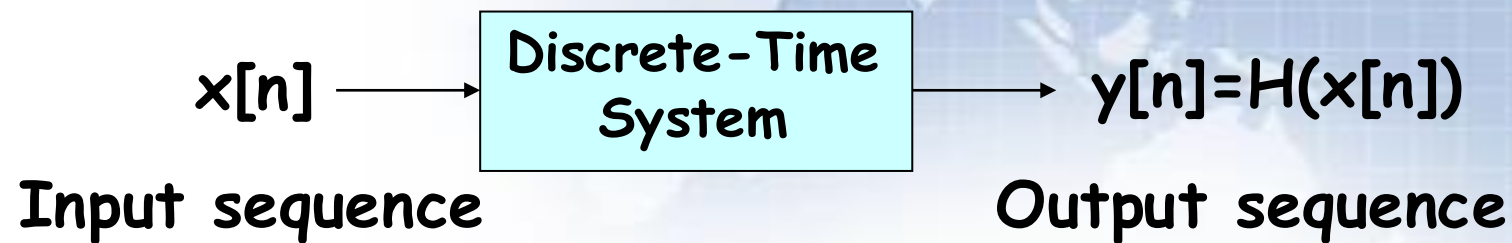
4

Phase and Group Delay



4.1 Discrete-Time Systems Examples

- A discrete-time system processes a given input sequence $x[n]$ to generate an output sequence $y[n]$ with more desirable properties
- In most applications, the discrete-time system is a single-input, single-output system:



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4.1 Discrete-Time Systems Examples



- Accumulator

$$y[n] = \sum_{l=-\infty}^{M-1} x[n-l]$$

In another form

$$y[n] = \sum_{l=-\infty}^{n-1} x[l] + x[n] = y[n-1] + x[n]$$

$$y[n] = \sum_{l=-\infty}^{-1} x[l] + \sum_{l=0}^n x[l] = y[-1] + \sum_{l=0}^n x[l]$$

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4.1 Discrete-Time Systems Examples



- **Moving-Average filter**

$$y[n] = \frac{1}{M} \sum_{l=0}^{M-1} x[n-l]$$

Exponentially Weighted Running Average filter

$$y[n] = \alpha y[n-1] + x[n]$$

Linear Interpolator

$$y[n] = x_u[n] + \frac{1}{2} (x_u[n-1] + x_u[n+1])$$

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4.1 Discrete-Time Systems Examples



- Median filter

$$y[n] = \text{med}\{x[n-K], \dots, x[n-1], x[n], x[n+1], \dots, x[n+K]\}$$

$$\text{med}\{2, -3, 10, 5, -1\} = 2$$

The median filter is implemented by sliding a window of odd length over the input sequence $\{x[n]\}$ one sample at a time.



4.2 Classification of Discrete-Time Systems

- **Linear System**
- **Shift-Invariant System**
- **Causal System**
- **Stable System**
- **Passive and Lossless Systems**



4.2 Classification of Discrete-Time Systems

- **Linear System**

- **Definition**

For an input signal $x[n] = \alpha x_1[n] + \beta x_2[n]$

The response is given by $y[n] = \alpha y_1[n] + \beta y_2[n]$

If superposition property hold for any **arbitrary** constants, α and β , and for **all possible input** signals ,the system can be called as **linear system**

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4.2 Classification of Discrete-Time Systems

Shift-Invariant System

- Definition

For an input signal

The response is given by

$$x_1[n] \Rightarrow y_1[n]$$

If

$$x[n] = x_1[n - n_0] \longrightarrow y[n] = y_1[n - n_0]$$

property hold for any n_0 can be given, the system is called as **shift-invariant system**

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4.2 Classification of Discrete-Time Systems

Causal System

Definition :

In addition to the above two properties, the n_0 th output sample $y[n_0]$ depends only on input samples $x[n]$ for $n \leq n_0$ and does not depend on input samples for $n > n_0$.

This system is called as **causal system**



4.2 Classification of Discrete-Time Systems

For a **causal system**

If $u_1[n] = u_2[n]$ for $n < N$

Implies also that $y_1[n] = y_2[n]$ for $n < N$

Note:

The definition of causality given above can be applied only to discrete-time systems with **the same sampling rate** for the input and the output.



4.2 Classification of Discrete-Time Systems

Stable Sytem

Definition: **If and only if**, for every bounded input, the output is also bounded, the system can be called as **stable system**

This type of stability is usually referred to as bounded-input, bounded-output (**BIBO**) staility.



4.2 Classification of Discrete-Time Systems

For a **BIBO system**, if the response to $x[n]$ is the sequence $y[n]$.

While $|x[n]| < B_x$, for all values of n

Then: $|y[n]| < B_y$ for all values of n

where B_x and B_y are **finite constants**.



4.2 Classification of Discrete-Time Systems

Passive and Lossless Systems

A discrete-time system is said to be **passive** if, for every finite energy input sequence $x[n]$, the output sequence $y[n]$ has, at most, the same energy, i.e.

$$\sum_{n=-\infty}^{\infty} |y[n]|^2 \leq \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

If the above inequality is satisfied with an equal sign for every input sequence, the discrete-time system is said to be **lossless**.



4.2 Classification of Discrete-Time Systems

The passivity and the losslessness properties are **crucial** to the design of discrete-time systems with very low **sensitivity to changes** in the filter coefficients



4.3 Impulse and Step Responses

Definition: The response of a digital system to a unit sample sequence $\{\delta[n]\}$ is called the **unit sample response**, or simply, the **impulse response**, and is denoted as $\{h[n]\}$.

The response of a discrete-time system to a unit step sequence $\{\mu[n]\}$, denoted as $\{s[n]\}$, is its unit step response or simply, **the step response**.

A linear time-invariant digital system can be completely **characterized** in the time-domain by its impulse response or its step response

4.4 Time-Domain Characterization of LTI Discrete-Time System



• Input-Output Relationship

Because

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

- The response of the LTI system to an input $x[k] \delta[n-k]$ will be $x[k] h[n-k]$

will be
$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

4.4 Time-Domain Characterization of LTI Discrete-Time System



- The summation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$ and represented compactly as

$$y[n] = x[n] \circledast h[n]$$

Convolution Sum



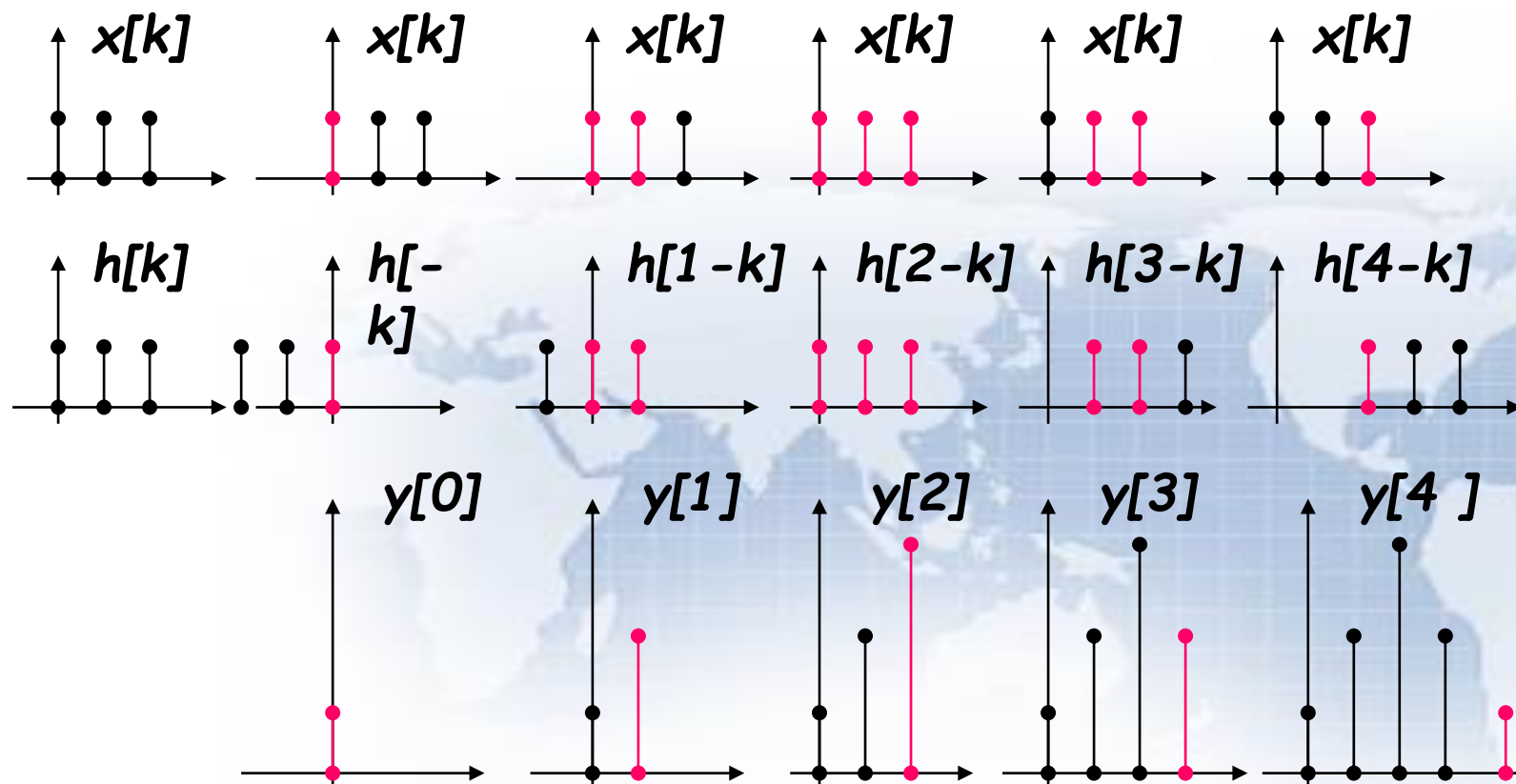
- Example: Develop the sequence $y[n]$ generated by the convolution of the sequences $x[n]$ and $h[n]$:

$$x[n] = h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

$$y[n] = x[n] \circledast h[n] = \sum_{k=0}^{\infty} x[k]h[n-k]$$

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$$x[n] = h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$



$$y[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 2\delta[n-3] + \delta[n-4]$$

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4.4 Time-Domain Characterization of LTI Discrete-Time System



Example $x[n] = u[n - N_1] - u[n - N_2 - 1]$

Length $N_2 - N_1 + 1$

$n: [N_1, N_2] \quad N_1 < N_2$

$h[n] = u[n - N_3] - u[n - N_4 - 1]$

Length $N_4 - N_3 + 1$

$n: [N_3, N_4] \quad N_3 < N_4$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

4.4 Time-Domain Characterization of LTI Discrete-Time System



$$y[n] = \begin{cases} \neq 0, & N_1 + N_3 \leq n \leq N_2 + N_4 \\ 0, & n > N_2 + N_4, \text{ or } n < N_1 + N_3 \end{cases}$$

Length $(N_2 + N_4) - (N_1 + N_3) + 1$

$$N_1 = N_3 = 0 \quad N_2 = 4, \quad N_4 = 3$$

So
$$y[n] = \begin{cases} \neq 0, & 0 \leq n \leq 7 \\ 0, & n > 7, \text{ or } n < 0 \end{cases}$$

Properties Of Convolution



1. Commutative property

$$y[n] = x[n] \circledast h[n] = h[n] \circledast x[n]$$

2. Associative property

$$\begin{aligned} x[n] \circledast h_1[n] \circledast h_2[n] &= [x[n] \circledast h_1[n]] \circledast h_2[n] \\ &= x[n] \circledast [h_1[n] \circledast h_2[n]] = [x[n] \circledast h_2[n]] \circledast h_1[n] \end{aligned}$$

3. Distributive property

$$\begin{aligned} x[n] \circledast [h_1[n] + h_2[n]] &= \\ x[n] \circledast h_1[n] + x[n] \circledast h_2[n] \end{aligned}$$

4.4 Time-Domain Characterization of LTI Discrete-Time System



- **Stability System Condition**

- If and only if the output sequence $y[n]$ of the system remains bounded for all bounded input sequence $x[n]$, a discrete-time is defined to be *BIBO* stable

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

4.4 Time-Domain Characterization of LTI Discrete-Time System



Causality System Condition

If and only if the response sequence $\{h[n]\}$ of a LTI discrete-Time system is satisfying the condition of Equation below, it is causal:

$$h[k]=0, \text{ for } k<0$$

Causality & Stability System Condition

$$h[n] = h[n]u[n] \quad \sum_{n=-\infty}^{\infty} |h[n]| = P < \infty$$

4.5 Simple Interconnection Schemes



1. Cascade connection

Definition: If the second LTI systems' input is connected to the first systems' output, we say the two system are connected in **cascade**. the overall impulse response $h[n]$ of the two cascaded-systems is given below:

$$h[n] = h_1[n] \otimes h_2[n]$$

(see Figure 4.8)

4.5 Simple Interconnection Schemes



2.Parallel connection

Definition: If the two LTI systems' output are added to form the new output while the same input is fed to both system, we say the two system are connected in **Parallel** (see Figure 4.9). The overall impulse response $h[n]$ is given below:

$$h[n] = h_1[n] + h_2[n]$$



4.6 Finite-dimensional LTI discrete-time system

Definition: If for the LTI system, the relation between input $x[n]$ and output $y[n]$ is given as below:

$$\sum_{k=0}^N d_k y[n-k] = \sum_{k=0}^M p_k x[n-k]$$

Here, d_k , p_k are constant. The order of the system is given by $\max(N, M)$.



4.6 Finite-dimensional LTI discrete-time system

If we assume the system to be causal and $d_0 \neq 0$, then we can rewrite the equation 4.32 as below:

$$y[n] = -\sum_{k=1}^N \frac{d_k}{d_0} y[n-k] + \sum_{k=0}^M \frac{p_k}{d_0} x[n-k] \quad (4.33)$$

Then, the output $y[n]$ can be computed for all $n \geq n_0$ knowing $x[n]$ and initial conditions $y[n_0-1]$, $y[n_0-2]$, ..., $y[n_0-N]$.



4.6 Finite-dimensional LTI discrete-time system

Total solution calculation

We may divided the answer of Eq(4.34) into two parts as below:

$$y[n] = y_c[n] + y_p[n]$$

Here , $y[n]$ is called **total solution** . $y_c[n]$ is the solution of Eq(4.34) with the input $x[n]=0$, which is called **complementary solution**. $y_p[n]$ is the solution of Eq(4.34) with the input $x[n] \neq 0$, which is called **particular solution**.

Self-Study



4.6 Finite-dimensional LTI discrete-time system

Zero-input response and zero-state response

We may also divided the answer of Eq(4.33) into the two parts as below:

$$y[n] = y_{zi}[n] + y_{zs}[n]$$

Here ,the **zero-input response $y_{zi}[n]$** is obtained by solving Eq(4.33) by setting the input $x[n]=0$,the **zero-state response $y_{zs}[n]$** is obtained by solving Eq(4.33) by applying the specified input with all initial conditions set to zero.

Self-Study



4.6 Finite-dimensional LTI discrete-time system

Example 4.32 Known the difference equation as bellow:

$$y[n] + y[n-1] - 6y[n-2] = x[n]$$

For a step input $x[n]=8\mu[n]$ and with initial conditions $y[-1]=1$ and $y[-2]=-1$, the total response $y[n]$ are wanted.

First, determine the **eigen value** of the difference equation

Setting $x[n]=0$ and $y[n]=\lambda^n$ in the difference equation we arrive at

Self-Study



4.6 Finite-dimensional LTI discrete-time system

$$\begin{aligned}\lambda^n + \lambda^{n-1} - 6\lambda^{n-2} &= \lambda^{n-2}(\lambda^2 + \lambda^1 - 6) \\ &= \lambda^{n-2}(\lambda + 3)(\lambda - 2) = 0\end{aligned}$$

And hence its **eigen value** (roots) are:

$$\lambda_1 = -3, \quad \lambda_2 = 2$$

Therefore we can get the **complementary solution** as below:

$$y_c[n] = \alpha_1(-3)^n + \alpha_2(2)^n$$

For the **particular solution** we assume

$$y_p[n] = \beta$$

Self-Study



4.6 Finite-dimensional LTI discrete-time system

The $y_p[n]$ is also satisfies the difference equation, so we arrive at:

$$\beta + \beta - 6\beta = 8\mu[n] \Rightarrow \beta = -2 \quad n \geq 0$$

The total response is the form

$$\begin{aligned} y[n] &= y_c[n] + y_p[n] \\ &= \alpha_1(-3)^n + \alpha_2(2)^n - 2 \quad n \geq 0 \end{aligned}$$

Consider the initial conditions we get

Self-Study



4.6 Finite-dimensional LTI discrete-time system

$$y[-2] = \alpha_1(-3)^{-2} + \alpha_2(2)^{-2} - 2 = -1$$

$$y[-1] = \alpha_1(-3)^{-1} + \alpha_2(2)^{-1} - 2 = 1$$

So we arrive at

$$\alpha_1 = -1.8, \quad \alpha_2 = 4.8$$

Thus

$$y[n] = -1.8(-3)^n + 4.8(2)^n - 2, \quad n \geq 0$$

Read and exercise example 4.23 and 4.24 by yourself!!

Self-Study



4.6 Finite-dimensional LTI discrete-time system

Impulse response calculation

We may get the **Impulse response** by set $x[n]=\delta[n]$, usually it has the form as $y_c[n]$

Example 4.25 Known the difference equation as bellow:

$$y[n] + y[n-1] - 6y[n-2] = x[n]$$

its $h[n]$ is wanted.

From example 4.22 we get

$$h[n] = \alpha_1(-3)^n + \alpha_2(2)^n, \quad n \geq 0$$

Self-Study



4.6 Finite-dimensional LTI discrete-time system

Substituting from the difference equation we get

$$h[0] = \alpha_1 + \alpha_2 = 1$$

$$h[1] + h[0] = -3\alpha_1 + 2\alpha_2 + \alpha_1 + \alpha_2 = 0$$

So $\alpha_1 = 0.6, \quad \alpha_2 = 0.4$

$$h[n] = 0.6(-3)^n + 0.4(2)^n, \quad n \geq 0$$

For any other input $x[n]$, we can get the **total solution** $y[n]$ according to formula as bellow:

$$y[n] = x[n] * h[n]$$

Self-Study



4.6 Finite-dimensional LTI discrete-time system

Impulse and Step Response computation Using Matlab

```
p=[0.8  -0.44  0.36  0.02];  
d=[1      0.7  -0.45  -0.6];  
[h,m] = impz(p,d,41);  
[s,m] = stepz(p,d,41);
```

Self-Study



4.7 Classification of LTI Discrete-time system

Based on impulse response length

If $h[n]$ is finite length, we call it **finite impulse response(FIR) system**, otherwise, we call it **infinite impulse response(IIR) system**.

Based on the output calculation process

If the output can be calculated just by present and past input the system is called **nonrecursive system**, otherwise is called **recursive system**.

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4.8 Frequency-Domain representations of LTI Discrete-Time systems



- Most discrete-time signals encountered in practice can be represented as a linear combination of sinusoidal discrete-time signals of different angular frequencies
- Thus, knowing the response of the LTI system to a single sinusoidal signal, we can determine its response to more complicated signals by making use of the superposition property

4.8 Frequency-Domain representations of LTI Discrete-Time systems

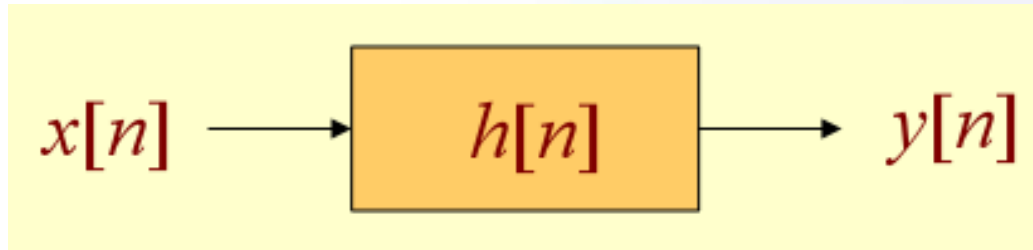


- An important property of an LTI system is that for certain types of input signals, called **eigen functions**, the output signal is the input signal multiplied by a complex constant
- We consider here one such eigen function as the input

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4.8.1 Frequency Response

- Consider the LTI discrete-time system with an impulse response $\{h[n]\}$ shown below



Its input-output relationship in the time-domain is given by the convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \quad (4.66)$$



4.8.1 Frequency Response

Definition

According to Eq(4.66),if we set

$$x[n] = e^{j\omega n}, \quad -\infty < n < \infty \quad (4.67)$$

The output of an LTI system is

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} \\ &= \left(\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} \\ &= H(e^{j\omega}) e^{j\omega n} \end{aligned} \quad (4.69)$$

Where $H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} \quad (4.70)$

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4.8.1 Frequency Response

Obviously , for a complex exponential input signal $e^{j\omega n}$, the output of an LTI discrete-time system is also the same signal multiplied by a complex constant $H(e^{j\omega})$.

Here , $e^{j\omega n}$ is called an **eigen function** of the LTI discrete-time system.

$H(e^{j\omega})$ is called the **frequency response** of the LTI discrete-time system , it provides a frequency-domain description of the system and is precisely the **DTFT of the impulse response $h[n]$** of the system.



4.8.1 Frequency Response

Obviously, that $H(e^{j\omega})$ can completely characterizes the LTI discrete-time system in the frequency domain.

Just like any other DTFT, $H(e^{j\omega})$ is usually also a complex function of ω with a period 2π and can be expressed in terms of its **real and imaginary** parts or its **magnitude and phase**.

$$H(e^{j\omega}) = H_{re}(e^{j\omega}) + jH_{im}(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)}$$

Where $\theta(\omega) = \arg\{H(e^{j\omega})\}$

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4.8.1 Frequency Response

The quantity $|H(e^{j\omega})|$ is called the **magnitude response** and the quantity $\theta(\omega)$ is called the **phase response** of the LTI discrete-time system.

The gain function:

$$G(\omega) = 20 \log_{10} |H(e^{j\omega})| \quad (dB)$$

Attenuation or loss function.

$$a(\omega) = -G(\omega) = 20 \log_{10} \frac{1}{|H(e^{j\omega})|}$$



4.8.1 Frequency Response

In engineering, **The gain function commonly use another definition as below:**

$$G(\omega) = 20 \log_{10} \frac{|H(e^{j\omega})|}{|H(e^{j\omega_0})|} \quad (dBC)$$

For a discrete-time system characterized by a **real** impulse response $h[n]$ (**real system**) should note:

1. $|H(e^{j\omega})|$ and $H_{re}(e^{j\omega})$ are even functions.
2. $\theta(\omega)$ and $H_{im}(e^{j\omega})$ are odd functions.

4.8.2 Frequency-Domain characterization

LTI Discrete-Time systems



If we applying the convolution theorem to the Eq.(4.66), we arrive at

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) \quad (4.74)$$

Where the $Y(e^{j\omega})$ and $X(e^{j\omega})$ denote the DTFT of $y[n]$ and $x[n]$.

So

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} \quad (4.75)$$

4.8.3 Frequency Response of LTI Discrete-Time systems



Frequency Response of LTI FIR Discrete-Time Systems

$$H(e^{j\omega}) = \sum_{k=N_1}^{N_2} h[k]e^{-j\omega k} \quad (4.77)$$

Frequency Response of LTI IIR Discrete-Time Systems

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M p_k e^{-j\omega k}}{\sum_{k=0}^N d_k e^{-j\omega k}} \quad (4.80)$$

4.8.4 Frequency Response Computation Using MATLAB



Example 4.31 For

$$h[n] = \begin{cases} \frac{1}{M}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases} \quad (4.81)$$

**The
frequency
response is
wanted.**

4.8.4 Frequency Response Computation Using MATLAB



Solution:

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{M} \sum_{n=0}^{M-1} e^{-j\omega n} = \frac{1}{M} \left(\sum_{n=0}^{\infty} e^{-j\omega n} - \sum_{n=M}^{\infty} e^{-j\omega n} \right) \\ &= \frac{1}{M} \left(\sum_{n=0}^{\infty} e^{-j\omega n} \right) (1 - e^{-jM\omega}) = \frac{1}{M} \frac{1 - e^{-jM\omega}}{1 - e^{-j\omega}} \\ &= \frac{1}{M} \frac{\sin(M\omega/2)}{\sin(\omega/2)} e^{-j(M-1)\omega/2} \end{aligned} \quad (4.82)$$

$$\left| H(e^{j\omega}) \right| = \left| \frac{1}{M} \frac{\sin(M\omega/2)}{\sin(\omega/2)} \right| \quad (4.83)$$

$$\theta(\omega) = -\frac{(M-1)\omega}{2} + \pi \sum_{k=1}^{\lfloor M/2 \rfloor} \mu\left(\omega - \frac{2\pi k}{M}\right) \quad (4.84)$$

Frequency Response Computation (Using MATLAB)



- The function **freqz(h,1,w)** can be used to determine the values of the frequency response vector **h** at a set of given frequency points **w**
- From **h**, the real and imaginary parts can be computed using the functions **real** and **imag**, and the magnitude and phase functions using the functions **abs** and **angle**

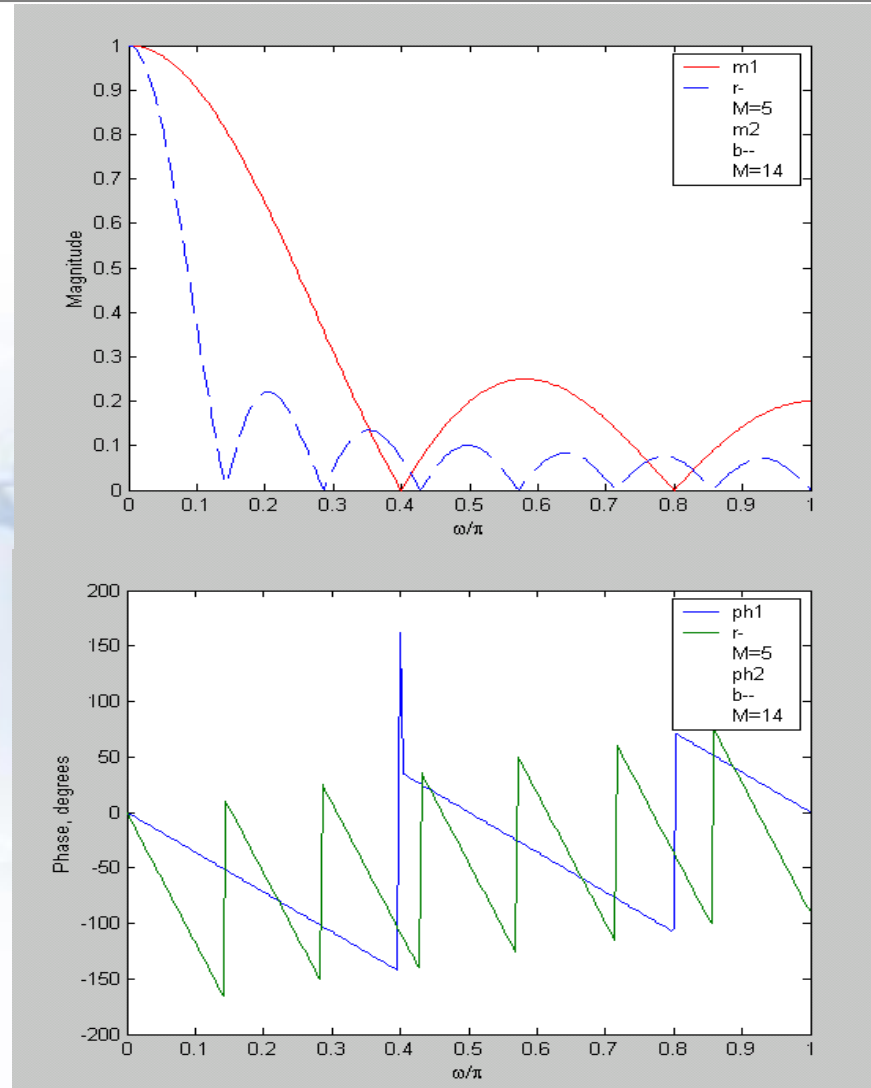
Program 4_3.m can be used to generate the magnitude and phase responses of an M-point moving average filter as shown on next-page

4.8.4 Frequency Response Computation Using MATLAB



**Magnitude
responses of the
moving-average
filters of length
5 and 14**

**Phase responses
of the moving-
average filters of
length 5 and 14**



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Frequency Response Computation (Using MATLAB)



- The phase response of a discrete-time system when determined by a computer may exhibit **jumps** by an amount 2π caused by the way the arctangent function is computed
- The phase response can be made a continuous function of ω by **unwrapping** the phase response across the jumps

Frequency Response Computation (Using MATLAB)



- To this end the function **unwrap** can be used, provided the computed phase is in radians
- The jumps by the amount of 2π should not be confused with the jumps caused by the zeros of the frequency response as indicated in the phase response of the moving average filter



4.8.5 Steady-state and transient Response

For a periodic signal $x[n]$, we should represent it as

$$x[n] = \sum_{l=-\infty}^{\infty} a_l e^{jl\omega_0 n}$$

So the response of an LTI system is

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k] x[n-k] \\ &= \sum_{k=-\infty}^{\infty} h[k] \sum_{l=-\infty}^{\infty} a_l e^{jl\omega_0 (n-k)} \end{aligned}$$



4.8.5 Steady-state and transient Response

$$= \sum_{l=-\infty}^{\infty} a_l \sum_{k=-\infty}^{\infty} h[k] e^{jl\omega_0(n-k)} = \sum_{l=-\infty}^{\infty} a_l H(e^{jl\omega_0}) e^{jl\omega_0 n}$$

That is shown, for **every frequency-part** of $x[n]$, the response of an LTI system is also a complex exponential signal of **the same frequency** multiplied by a complex constant $H(e^{jl\omega_0})$.

The response of an LTI system of a periodic input is commonly called the **steady-state response**.

The transient Frequency response of the LTI system:

$$H_n(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$$



4.8.6 Response to a Causal Exponential Sequence

While the input of the LTI system is a causal exponential sequence :

$$x[n] = e^{j\omega n} \mu[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} \mu(n-k)$$

$$= \left(\sum_{k=-\infty}^n h[k] e^{-j\omega k} \right) e^{j\omega n}$$

$$= H_n(e^{j\omega}) e^{j\omega n}$$



4.8.6 Response to a Causal Exponential Sequence

$$\begin{aligned} &= \left(\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} - \left(\sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} \\ &= H(e^{j\omega}) e^{j\omega n} - \left(\sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} \\ &= y_{sr}[n] + y_{tr}[n] \end{aligned}$$

**The steady-state
response:**

$$y_{sr}[n] = H(e^{j\omega}) e^{j\omega n}$$

**The transient
response:**

$$y_{tr}[n] = - \left(\sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n}$$

4.8.7 Concept of Filtering



Concept of Filtering

One application of an LTI discrete-time system is to **pass certain frequency components** in an input sequence without any distortion (if possible) and to **block other frequency components**.

Such systems are called **digital filters** and are one of the main subjects of discussion in this text.

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4.8.7 Concept of Filtering



- The key to the filtering process is

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

- It expresses an arbitrary input as a linear weighted sum of an infinite number of exponential sequences, or equivalently, as a linear weighted sum of sinusoidal sequences

MOOC

4.8.7 Concept of Filtering



- Thus, by appropriately choosing the values of the magnitude function $|H(e^{j\omega})|$ of the LTI digital filter at frequencies corresponding to the frequencies of the sinusoidal components of the input, some of these components can be selectively heavily attenuated or filtered with respect to the others

Example (4.33) known

$$\begin{aligned}x[n] &= x_1[n] + x_2[n] \\ &= \{\cos(0.1n) + \cos(0.4n)\}\mu[n]\end{aligned}$$

MOOC

4.8.7 Concept of Filtering



Design a Filter to pass the high-frequency component($x_2[n]$) but block the low-frequency component($x_1[n]$) of $x[n]$.

Solution :For simplicity we assume the filter to be an FIR filter of length 3 with an impulse response :

$$\begin{aligned}h[0] &= h[2] = \alpha, \quad h[1] = \beta \\ \therefore H(e^{j\omega}) &= h[0] + h[1]e^{-j\omega} + h[2]e^{-j2\omega} \\ &= \alpha + \beta e^{-j\omega} + \alpha e^{-j2\omega} \\ &= (2\alpha \cos \omega + \beta)e^{-j\omega}\end{aligned}\tag{4.93}$$

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4.8.7 Concept of Filtering



**The magnitude
function**

$$|H(e^{j\omega})| = |2\alpha \cos \omega + \beta| \quad (4.94)$$

**The phase
function**

$$\theta(\omega) = -\omega \quad (4.95)$$

In order to stop the low-frequency component from appearing at the output of the filter, the magnitude function at $\omega=0.1$ should be equal to zero. Similarly, the magnitude function at $\omega=0.4$ must equal to 1. So

$$\left. \begin{array}{l} 2\alpha \cos(0.1) + \beta = 0 \\ 2\alpha \cos(0.4) + \beta = 1 \end{array} \right\} \begin{array}{l} \alpha = -6.76195 \\ \beta = 13.456335 \end{array}$$

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4.8.7 Concept of Filtering



- Thus the output-input relation of the FIR filter is given by

$$y[n] = -6.76195(x[n] + x[n-2]) + 13.456335x[n-1]$$

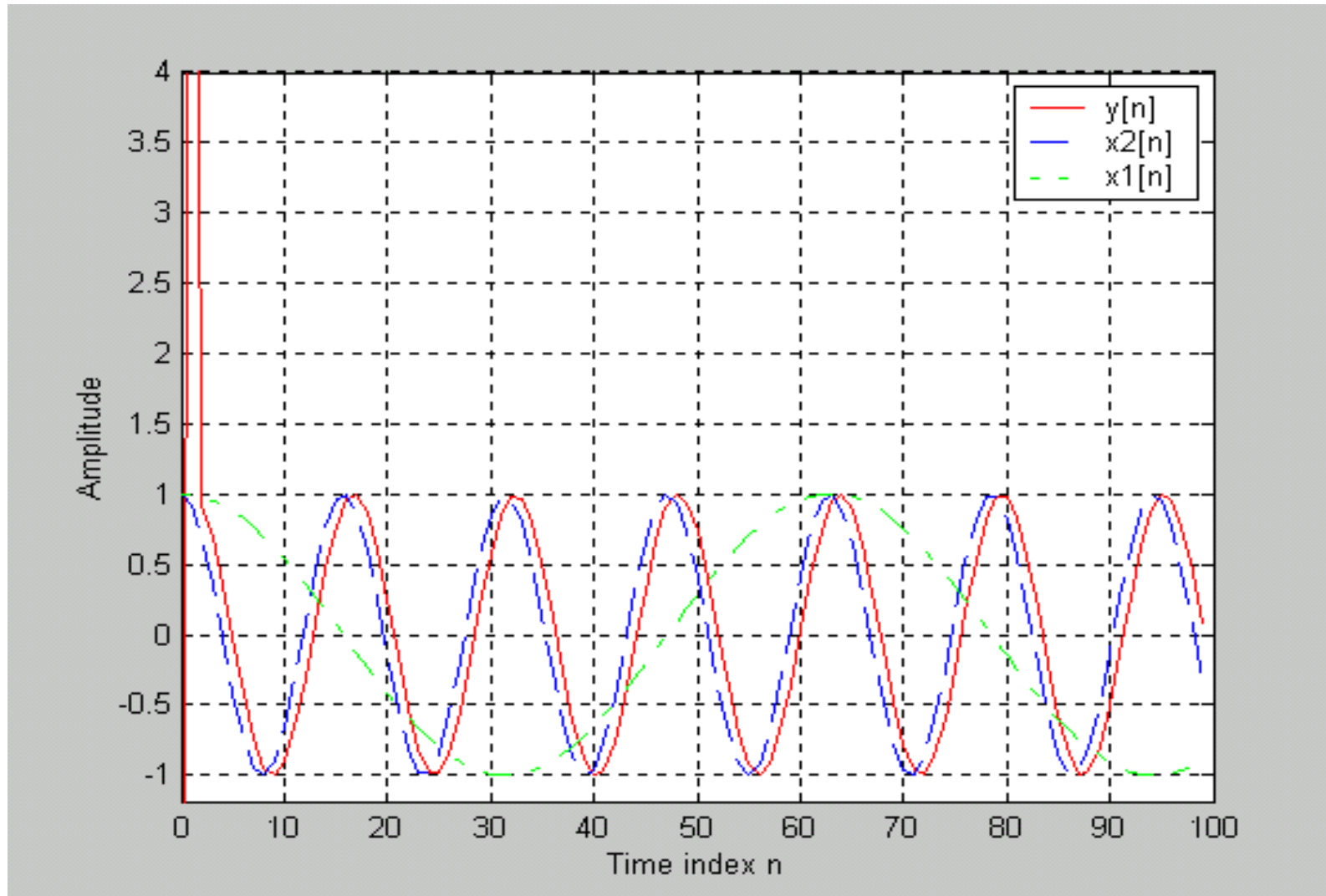
where the input is

$$x[n] = \{\cos(0.1n) + \cos(0.4n)\}\mu[n]$$

- Program 4_4.m can be used to verify the filtering action of the above system

MOOC

4.8.7 Concept of Filtering



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4.8.7 Concept of Filtering



- The first seven samples of the output are shown below

n	$\cos(0.1n)$	$\cos(0.4n)$	$x[n]$	$y[n]$
0	1.0	1.0	2.0	-13.52390
1	0.9950041	0.9210609	1.9160652	13.956333
2	0.9800665	0.6967067	1.6767733	0.9210616
3	0.9553364	0.3623577	1.3176942	0.6967064
4	0.9210609	-0.0291995	0.8918614	0.3623572
5	0.8775825	-0.4161468	0.4614357	-0.0292002
6	0.8253356	-0.7373937	0.0879419	-0.4161467

MOOC



4.8.7 Concept of Filtering

- From this table, it can be seen that, neglecting the least significant digit,

$$y[n] = \cos(0.4(n-1)) \quad \text{for } n \geq 2$$

- Computation of the present value of the output requires the knowledge of the present and two previous input samples
 - Hence, the first two output samples, $y[0]$ and $y[1]$, are the result of assumed zero input sample values at $n = -1$ and $n = -2$

4.8.7 *Concept of Filtering*



- Therefore, first two output samples constitute the transient part of the output
- Since the impulse response is of length 3, the steady-state is reached at $n = N = 2$
- Note also that the output is delayed version of the high-frequency component $\cos(0.4n)$ of the input, and the delay is one sample period



4.9 Phase and Group Delays

Definitions: If the input is a sinusoidal signal of frequency ω_0 but lagging in phase by $\theta(\omega_0)$ radians, as demonstrated in Eq.(4.86):

$$\begin{aligned}x[n] &= A \cos(\omega_0 n + \phi), \quad -\infty < n < \infty \\ \therefore x[n] &= \frac{A}{2} \{e^{j\phi} e^{j\omega_0 n} + e^{-j\phi} e^{-j\omega_0 n}\} \\ \therefore y[n] &= \frac{A}{2} \{e^{j\phi} H(e^{j\omega_0}) e^{j\omega_0 n} + e^{-j\phi} H(e^{-j\omega_0}) e^{-j\omega_0 n}\} \\ &= \frac{A}{2} |H(e^{j\omega_0})| \{e^{j\phi} e^{j\theta(\omega_0)} e^{j\omega_0 n} + e^{-j\phi} e^{-j\theta(\omega_0)} e^{-j\omega_0 n}\} \\ &= A |H(e^{j\omega_0})| \cos(\omega_0 n + \theta(\omega_0) + \phi)\end{aligned}$$



4.9 Phase and Group Delays

From example 4.32 we can see that the output signal $y[n]$ of an LTI discrete-time system exhibits some delay relative to the input signal $x[n]$ caused by the nonzero phase response of the system.

We can rewrite Eq.(4.86) as

$$y[n] = A |H(e^{j\omega_0})| \cos(\omega_0(n + \frac{\theta(\omega_0)}{\omega_0}) + \phi) \quad (4.99)$$

indicating a time delay, more commonly known as **phase delay** at $\omega = \omega_0$ given by

$$\tau_p(\omega_0) = -\frac{\theta(\omega_0)}{\omega_0} \quad (4.100)$$



4.9 Phase and Group Delays

While input is

$$x[n] = A \cos(\omega_0 n) \cos(\omega_c n)$$

Where $\omega_c \gg \omega_0$ $x[n]$ is a narrowband signal .

In this situation ,we can derivate the output signal as below:

$$y[n] \approx A \left| H(e^{j\omega_c}) \right| \cos(\omega_0 (n - \tau_g)) \cos(\omega_c (n - \tau_p))$$

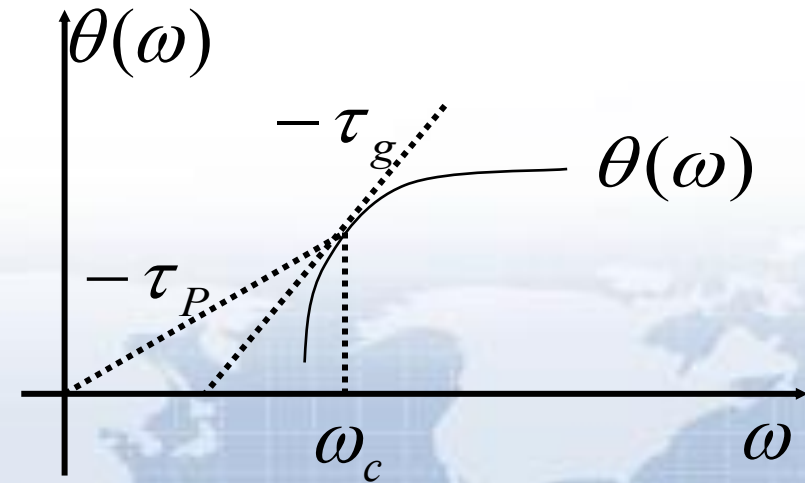
**group delay or
envelope delay**

$$\tau_g = - \left. \frac{d\theta(\omega)}{d\omega} \right|_{\omega=\omega_c} \quad (4.101)$$

4.9 Phase and Group Delays

phase delay $\tau_p = -\frac{\theta(\omega_c)}{\omega_c}$

An example of **group delay** and **phase delay** is shown in Figure 4.15

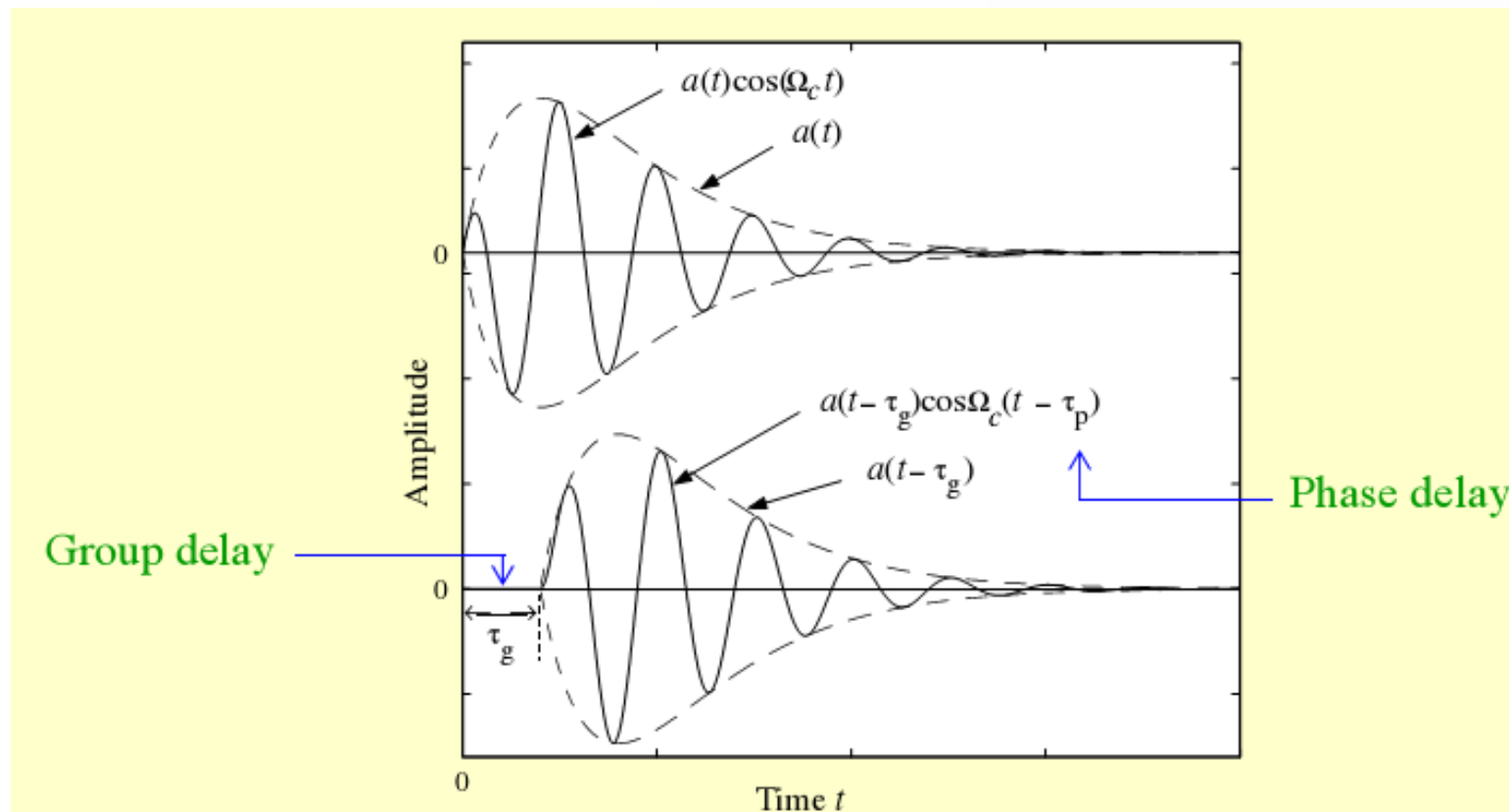


The waveform of the underlying continuous-time output shows **distortion** when the τ_g of the LTI system is not constant over the bandwidth of $x[n]$. A **delay equalizer** is usually cascaded with the LTI system

4.9 Phase and Group Delays



- The figure(4.16) below illustrates the effects of the two delays on an amplitude modulated sinusoidal signal



4.10 Phase Delay and Group Delay Computation using MATLAB



Read and exercise by yourself !!

- Phase delay can be computed using the function **phasedelay**
- Group delay can be computed using the function **grpdelay**

Reference Homework



- Read textbook from p277 to 319

- Problems

4.3, 4.4, 4.5, 4.8, 4.10, 4.12, 4.23, 4.30, 4.35, 4.61, 4.62, 4.69

M4.1, M4.2, M4.5